Data package / Supplementary Material S7

Hand-loading observations

Complete manual and synchronized results, with the static controls and exploratory ramp figure.

Scope and setup

Series labels are synchronized presses (S), static controls on the same assembly (SC), exploratory ramps (SX), and held-out test (B).

The manual scale readings were approximate; their temporal correspondence to the Hall averaging windows was not independently verified. In the synchronized assembly the Hall head was not mechanically fastened to the load-cell platform. A later setup check showed support deflection under load, which was not quantified during the recorded trials. The indication differences therefore are not used to quantify the force accuracy of the head itself or identify a friction mechanism.

The centered manual series used a CRW-7024 zeroed with the 343 g Hall assembly in place. The synchronized series placed the head on a 111 g load-cell platform and applied hand force to the Hall tray. The `setup.description` in the preserved plan export specifies intended taped mounting; it does not describe the actual unfastened assembly. Recording notes describe the actual arrangement.

The S synchronized-press analysis rules were fixed before acquisition. SC was added after the S analysis, with its rules fixed before SC acquisition. SX had a separate exploratory plan from the outset. The later time-shift sensitivity is distinguished below. The separate 15-press directional-contrast experiment remains in manuscript Figure 4 and Supplementary Material S7; it is not the S series tabulated here.

Manual scale readings: all 18 pairs

For each press, the last complete 100-sample device window ending before the reading cue was paired with the entered scale indication. All 15 initial presses and three follow-up presses are shown. The follow-up used a new display zero and the same calibration. No synchronized scale stream or video was recorded. The difference column uses the stored unrounded readings, so rounded gram columns may not reproduce its last digit.

Table 1. Manual indication pairs.

Press	Entered scale (g)	Recorded device (g)	Device minus scale (N)
Initial 1	532	526	-0.06
Initial 2	975	945	-0.29
Initial 3	2010	1839	-1.68
Initial 4	3020	2407	-6.01
Initial 5	3990	3893	-0.96
Initial 6	2020	1893	-1.25
Initial 7	4010	3933	-0.76
Initial 8	980	1022	+0.41
Initial 9	3010	2943	-0.65
Initial 10	514	541	+0.27
Initial 11	3030	2631	-3.91
Initial 12	512	483	-0.28
Initial 13	4010	3931	-0.78
Initial 14	2010	2013	+0.03
Initial 15	1005	988	-0.16
Follow-up 1	3060	2794	-2.61
Follow-up 2	3040	2796	-2.40
Follow-up 3	2950	2804	-1.43

Sources: initial series and follow-up series. The records include window timing, scale entries and device output checks.

Synchronized presses: all 18 presses

Hall and load-cell device clocks were mapped to host UTC by linear regression. A press was detected above 0.3 kg for at least 1.5 s. Release was the last sample at or above 90% of the median force over the middle half of that interval. The analysis window was the final 1 s before release; a 3 s window was also evaluated. Onset was the last sample at or below 0.05 kg preceding the interval. Both systems used the 3 s ending 0.2 s before onset as unloaded reference. Hall indications used the frozen four-channel interpolation. These rules are in the preserved plan export.

The later timing sensitivity shifted the Hall windows by the median edge lag (-0.18 s, Hall minus load-cell edge time). Each edge was the rising 50% crossing of that system's own plateau. The lag-shifted column remains a sensitivity calculation; it does not replace the original window result.

Table 2. Synchronized calibrated indications and contrast magnitude.

Round	Application	Load cell (N)	Hall (N)	Hall minus load cell (N)	Lag-shifted difference (N)	Q (ADC counts)
1	Centered	24.32	21.15	-3.17	-2.22	72.0
1	Tilted	25.85	20.51	-5.34	-5.32	73.7
1	15 mm offset	24.73	19.90	-4.83	-4.79	89.9
2	Centered	24.03	21.56	-2.47	-1.94	79.5
2	Tilted	25.41	21.33	-4.08	-3.53	71.6
2	15 mm offset	25.00	20.00	-5.00	-4.96	78.5
3	Centered	24.97	22.02	-2.96	-2.80	81.3
3	Tilted	24.75	22.51	-2.24	-1.82	81.8
3	15 mm offset	24.40	20.11	-4.30	-4.29	86.8
4	Centered	25.80	21.71	-4.09	-3.60	72.5
4	Tilted	24.63	21.50	-3.13	-2.83	82.6
4	15 mm offset	25.02	20.00	-5.02	-5.01	76.5
5	Centered	24.52	21.69	-2.83	-2.33	67.1
5	Tilted	25.20	23.37	-1.82	-1.25	61.2
5	15 mm offset	24.87	21.60	-3.27	-3.09	91.1
1-3 kg sequence	Centered, 1 kg	10.25	9.86	-0.39	-0.08	33.3
1-3 kg sequence	Centered, 2 kg	19.81	16.34	-3.46	-3.43	53.7
1-3 kg sequence	Centered, 3 kg	29.61	23.94	-5.67	-5.68	83.6

Q is the magnitude of the two signed channel contrasts. Stored one-second, three-second and shifted-window values are in synchronized_press.json; reproduce_synchronized.py reconstructs them.

Static controls on the synchronized assembly

Each loaded record used its preceding unloaded record and the mean over 10-20 s of device time. Indication minus known mass times 0.00980665 N/g is listed for both systems. The four nearest-mass independent B placements are retained below for comparison; they used a different test arrangement and are not matched repetitions of SC.

Table 3. All four SC placements and the previously listed B comparisons.

Record	Mass (g)	Load cell minus applied load (N)	Hall minus applied load (N)
SC_1000_A	1000	-0.008	-0.50
SC_1000_B	1000	+0.003	-0.60
SC_2003_A	2003	-0.072	-1.19
SC_2003_B	2003	-0.073	-1.45
Independent test: B_T1_WATER_1100_C	1100		-0.47
Independent test: B_T2_WATER_1100_C	1100		-0.84
Independent test: B_T1_WATER_1850_C	1857		-1.86
Independent test: B_T2_WATER_1850_C	1857		-0.34

Separate exploratory ramps

Two 90 s records each contained two slow centered loading and unloading ramps, approximately 0 to 3 kg in 10 s, a 2 s hold, and return to zero in 10 s. Both systems were referenced to the first 3 s of each record. At each load-cell sample, the Hall indication was averaged over +/-0.05 s around the time shifted by the median S lag. Loading, unloading and hold branches used the gradient of the five-sample smoothed load-cell force; an absolute gradient below 0.2 N/s was classified as hold.

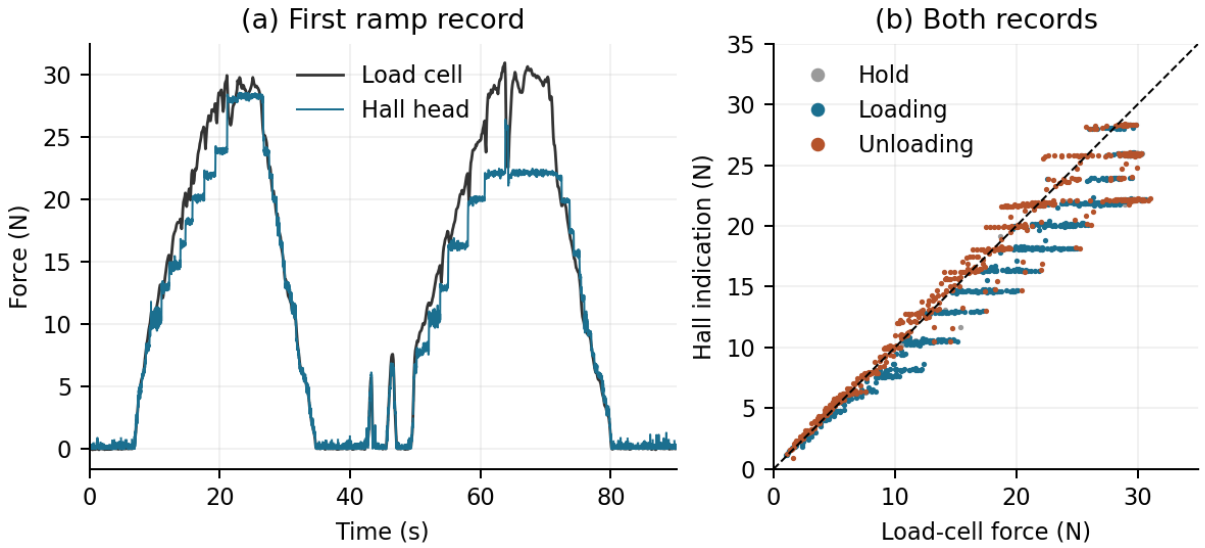


Figure 1. Left: the first ramp record. Right: paired indications from both records, with branches shown separately. The dashed line denotes equal indications. These traces describe the two recorded indications on the combined assembly; they do not isolate the contribution of the head or support.

The stored summaries are in synchronized_press.json. Reproduction: `python reproduce_synchronized.py`. Figure regeneration: `python plot_synchronized_ramps.py -output-dir figures`.

Record links and analysis history

The raw index maps the capture IDs below to the compressed measurement files. The data dictionary defines the timestamp and channel fields. Initial manual capture: `66c0b8ea13c5411b95ba2c51c7ca6e9c`; follow-up capture: `15a216a084594cd28d0df6cc6a5b0010`.

Capture pairs used by the stored S, SC and SX analyses.

Record	Hall capture	Load-cell capture
S_R1	506ceb3efdfc43299e3caf23c8bc920d	ba3f7c32724b465aa7df4030e8cb45e5
S_R2	e08d6454cb654d4299e838f585a49393	7b8a51a4e3024f9db961793483123506
S_R3	c7601aa6e7674e7aa2270baede14f692	952f83cf420843b6a9973ef487a1c326
S_R4	7a4adea027cd4e8eaaed7a759ca496f7	45641906f39c40b8ac451f5e7da78aca
S_R5	f29882bbd2064530ab01ea60c5437ecb	1b992053f1264d7aae0327fbbf3d45b0
S_LADDER	246bb06282714f088e0d58df6d1e6325	701812a5f05c48b0b6c4d86bd10c2906
SC_Z0	f1283d1c78704d3fa7515264b8833344	a948825a5b034fc4bfaf0b685daa9208
SC_1000_A	01a1d3e033054337b5bc26a7d2552c04	28546d6a0b0a43a0816f2a9c11748c41
SC_Z1	c5a426a954b846cea410e8601dbce883	03072da775fe473582393eef0c30837e
SC_1000_B	ea6b9302b4884e819f8731aa7c097afe	91c757b606e74075b05ce036f722c5f0
SC_Z2	277ae666240a476d92aefe273e89c1ab	957aede6436a4677b931312ace7e26c7
SC_2003_A	8934d20e737940db9239930b8a011601	b529bd563c1249c6981c6176f6358342
SC_Z3	e66f335de1b34d84ab6797aa9e92d735	d4ba3289012244e9830a0388b4e65d2c
SC_2003_B	24b0d1c997b14092ae0016a2be517007	60332ed81d8c498f9549205931122cc1
SX_RAMP1	d03586e7f3d84b03a197c5bf31fc55e6	759f91bc7ebf416bb2d4c27e06b52fdc
SX_RAMP2	15a9eaab9aa34d259fc66848981f751f	64a6e029bcbe45009c62bc846de2ff2a

The original frozen plan hashes are retained in synchronized_press_plan.json. The S, SC and SX roles are distinct:

- **S:** development_validation; analysis plan fixed before acquisition
- **SC:** control added after the S analysis; analysis fixed before acquisition
- **SX:** exploratory; reported descriptively

This report formats the stored numerical results without refitting models, choosing new windows or excluding rows. To check its rendering against those values, run `python build_hand_loading_report.py -check`.